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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x \sin^2 x}{(x \cos x - \sin x)} &= \frac{0}{0} \\ &\stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{(x \sin^2 x)'}{(x \cos x - \sin x)'} \\ &= \lim_{x \rightarrow 0} \frac{(\sin^2 x + 2x \sin x \cos x)}{(\cos x - x \sin x - \cos x)} \\ &= \lim_{x \rightarrow 0} \frac{\sin x (\sin x + 2x \cos x)}{-x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{(\sin x + 2x \cos x)}{-x} = \frac{0}{0} \\ &\stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{(\sin x + 2x \cos x)'}{(-x)'} \\ &= \lim_{x \rightarrow 0} \frac{(\cos x + 2 \cos - 2x \sin x)}{-1} \\ &= -(1 + 2 - 0) = -3\end{aligned}$$

References